Probability: Random experiment, repeated infinite number of times under the same conditions

All outcomes are known and well defined

def sample space: set of all possible outcomes S = sample space

Rolling dice: $S = \{1, 2, 3, 4, 5, 6\}$ → finite sample space

coin Toss : $S = \{H, T\}$ 3 coins : $|S| = 2^3$

Toss coin until head occurs: $S = \{H, TH, TTH \ldots\}$ discrete sample space (countable)

$S = \mathbb{R}$ continuous sample space (uncountable)

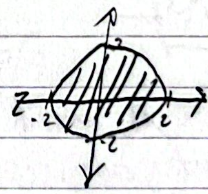
An event A is a subset of sample space $A \subset S$

Rolling dice $S = \{1, 2, 3, 4, 5, 6\}$

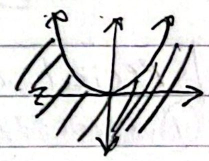
A = even # $A = \{2, 4, 6\}$

Ex $S = \mathbb{R}^2$ $S = \{(x, y) \mid x \in \mathbb{R}, y \in \mathbb{R}\}$

$A = \{(x, y) \mid x^2 + y^2 \leq 4\}$



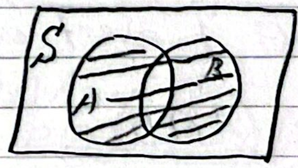
$B = \{(x, y) \mid y \leq x^2\}$



$\boxed{\text{Algebra of events}}$

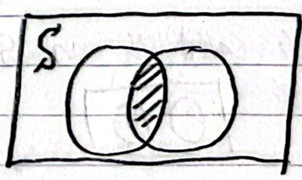
union, intersection, compliment

def A union of 2 events A, B is an event whose outcome belongs to A or B



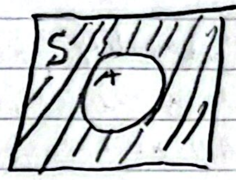
$A \cup B$
union

An intersection A, B is an event whose outcome is in both A and B



$A \cap B$
intersection

Complement: every element in the sample space not in the event

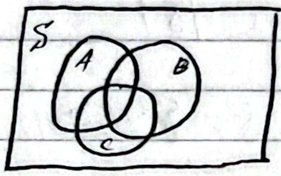


A', A^c

A null event $\emptyset \subset S$ (empty set)

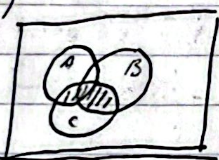
Intersection & union of more than 2 events

A, B, C

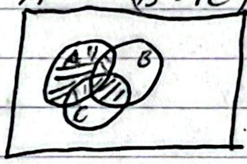


$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$

$(A \cup B) \cap C$



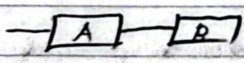
$A \cup (B \cap C)$



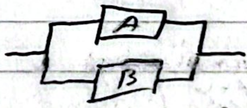
De Morgan Rule

$(A \cup B)^c = A^c \cap B^c$ $\quad$ $(A \cap B)^c = A^c \cup B^c$

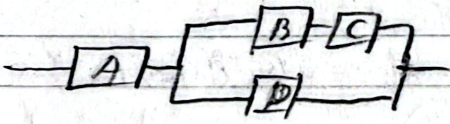
Ex



$A \cap B$



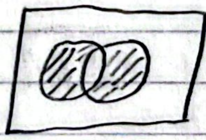
$A \cup B$



$A \cap ((B \cap C) \cup D)$

Special cases A and B are events

① Exactly 1 event occurs

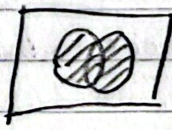


$(A \cup B) \cap (A \cap B)^c$

② At most 1 event occurs

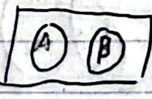
$(A \cap B)^c$

③ At Least 1



$A \cup B$

Def 2 events A and B in S are mutually exclusive (disjoint) if $A \cap B = \emptyset$



Probability function

Def: The function P from the set of all events defined on S is called a probability function: $\forall A$, $P(A) \geq 0$

Mutually exclusive = $\forall A \& B$ $A \cap B = \emptyset$

S is finite $\rightarrow$ $S = \{S_1, S_2, \ldots, S_n\}$ All outcomes are equally likely

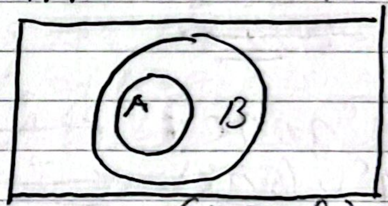
$P(S_1) = P(S_2) = \ldots = P(S_n)$

$P(A) = \frac{|A|}{|S|} = \frac{k}{n}$ $A = \{S_{i_1}, S_{i_2}, \ldots S_{i_k}\}$ $A \subset S$

Ex. Die roll, Event A = even # $\frac{|A|}{|S|} = \frac{3}{6} = 1/2$

$\boxed{\text{Properties of } P}$ Thm 1: $P(A^c) = 1 - P(A)$

Thm 2: $P(\emptyset) = 0$ $\emptyset = S^c$ Thm 3: If $A \subset B$ $P(A) \leq P(B)$



Thm 4: Let $A_1, A_2, \ldots A_n$ be mutually exclusive events $A_i \cap A_j = \emptyset$ $i \neq j$

$(B \cap A^c) \cup A = B$

Then $P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i)$ $A \cap \underbrace{(B \cap A^c)} = \emptyset$

Mutually exclusive

proof:

Base $P(A_1 \cup A_2) = P(A_1) + P(A_2)$ if $A_1 \cap A_2 = \emptyset$

Assume $A_1, A_2, \ldots A_n$ are mutually excusive and $P(A_1 \cup A_2 \ldots \cup A_n) = P(A_1) + P(A_2) + \ldots + P(A_n)$

let $B = P(A_1 \cup \ldots \cup A_n)$ then $B \cap A_{n+1} = \emptyset$

Then $P(A_1 \cup A_2 \cup \ldots \cup A_n \cup A_{n+1}) = P(A_1) + \ldots + P(A_n) + P(A_{n+1})$

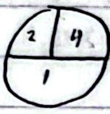
Thm 5: Let A and B be events, then $\underbrace{P(A) \cup P(B)}_{\text{Mutually exclusive}} = P(A) + P(B) - P(A \cap B)$

Thm 6: Let A, B, C be events $P(A \cup B \cup C) =$

$= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

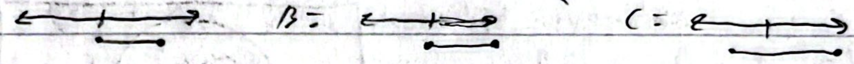
2.3.11 65% LL 70% ?L $70-65=5\%$
90% L? $90-65=25\%$
$+ = \boxed{30\%}$

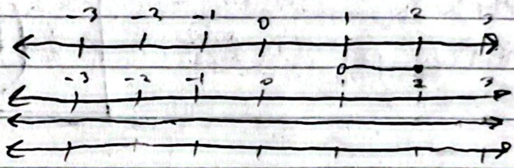
practice problems

2.2.10  U = first dart V = second dart
$S=\{(1,1),(1,2),(2,1),(1,4),(4,1),(2,4),(4,2),(2,2),(4,4)\}$

$S(\text{sums}) = \{2,3,4,5,6,8\}$

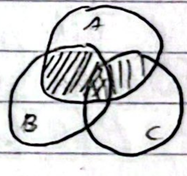
2.2.8 chances of walking on the 6th pitch
$S=\{(S,S,B,B,B,B),(S,B,S,B,B,B),(S,B,B,S,B,B),$
$(S,B,B,B,S,B),(B,S,S,B,B,B),(B,S,B,S,B,B),(B,S,B,B,S,B)$
$(B,B,S,S,B,B),(B,B,S,B,S,B),(B,B,B,S,S,B)\}$

2.2.20 $A=\{x: 0\le x\le 1\}$, $B=\{x: 0\le x\le 3\}$ $C=\{x: -1\le x\le 2\}$
A = B = C =


a. $A^c \cap B \cap C =$
b. $A^c \cup (B\cap C) =$ $\mathbb{R}$
c. $A\cap B\cap C^c =$ nothing $\varnothing$


2.2.33 $A\cap(B\cup C) = (A\cap B)\cup(A\cap C)$

$A\cup(B\cap C) = (A\cup B)\cap(A\cup C)$

2.3.4 A, B defined on S A, B, AB = 0.2 $AB' = 0.1$
$P(B) = 0.2$

$1-0.65 = 0.35$ At least 1

☆ 2.3.11 $0.9\cdot 0.3 + 0.1\cdot 0.7$ $0.07 + 0.27 = \boxed{0.34}$

$(A\cap B^c)\cup(A^c\cap B)$

2.3.16 $A = d_1+d_2 = 6$ $B = d_1 = d_2$
$A=\{(1,5),(5,1),(2,4),(4,2),(3,3)\}$
$B=\{(1,2),(2,1),(2,4),(4,2),(3,6)(6,3)\}$
$P(A\cap B^c) = \{(1,5),(5,1),(3,3)\} = \boxed{1/12}$